

LEON3 Ethernet test code documentation and execution guide ▾

greth_test.c

C

i have uploaded a c file for a etherent test for leon3 processor that runs in vcs on a linux terminal...can u me a document that explains what the c code is doing and how to run it and see it's working

Viewed 2 files >

Now let me read the docx skill and create the document.

Reading docx skill >

Now I have everything I need. Let me write the document generation script.

Ran 3 commands, created a file, read a file >

Here's your Word document! It covers everything in 9 sections:

1. **Overview** — what the program is and how it self-checks
2. **Hardware & Memory Map** — all register addresses, the GRLIB report mechanism, and DMA descriptor layout
3. **Code Structure & Function Reference** — a table of every function and what it does
4. **Test-by-Test Explanation** — detailed breakdown of all 5 tests including exact step sequences for Tests 4 & 5
5. **How to Build and Run** — prerequisites, the 4 steps (copy → compile → vcs-launch → read transcript), and common make targets
6. **Expected VCS Transcript Output** — the exact console output you should see when everything passes (22 checks)
7. **Troubleshooting** — 7 common failure symptoms with causes and fixes
8. **Key Configurable Constants** — FRAME_SIZE, NUM_FRAMES, TIMEOUT, PHY_ADDR
9. **Pass/Fail Summary** — what each test's PASS actually proves at the hardware level



Greth test documentation

this test will work even if the ethernet is taken outside the leon3 processor ryt? or will that be a different testing code?

Great question. The short answer is: **this specific code will NOT work outside the LEON3 processor** — you would need a different testing approach. Here's why:

Why this code is LEON3-specific

1. Bare-metal memory-mapped register access The entire test works by casting hardcoded addresses like `0x80000D00` to pointers and reading/writing them directly. This only works when your code is *running on the LEON3 itself*, where the CPU's memory bus is physically connected to those APB/AHB registers.
2. The addresses are from LEON3's memory map `0x80000D00` (GRETH), `0x80000200` (GRLIB testbench report) — these are LEON3 system addresses defined in `config.vhd`. A CPU outside the SoC has no visibility into these addresses at all.
3. The DMA descriptors live in LEON3's SRAM Tests 4 & 5 write descriptor structs into LEON3 local memory and pass their addresses to GRETH's AHB DMA. An external tester can't do this — it has no access to that memory space.
4. The compiler target is SPARC `sparc-gaisler-elf-gcc` produces SPARC V8 machine code. It won't run on an x86 Linux host or any external microcontroller.

What "external" testing would look like

It depends on *what* is external:

Scenario	Approach
Test from a Linux PC over real Ethernet	Send/receive raw Ethernet frames using <code>libpcap</code> or raw sockets (<code>AF_PACKET</code>) on the PC side; the LEON3 runs a simple loopback firmware
Test from another FPGA/board	A second device sends known frames; you capture and verify them on the receiving end
Test the GRETH IP core in isolation (without LEON3)	Write a pure VHDL/SystemVerilog testbench that drives the AHB/APB buses directly — no CPU involved at all

Scenario	Approach
Use GRETH's built-in EDCL debug link	GRETH has an Ethernet Debug Communication Link (EDCL) that lets a host PC talk to LEON3 memory over UDP — but this still requires LEON3 to be running

Bottom line

This code is tightly coupled to:

- The **LEON3 CPU** executing it
- The **GRLIB SoC memory map**
- The **VCS + GRLIB testbench** infrastructure

If you want to test GRETH from *outside* the processor, you'd need either a **VHDL-level testbench** (drives the bus signals directly) or a **host-side packet tool** (libpcap/raw sockets) paired with minimal firmware still running on LEON3. Want me to sketch out what either of those would look like?

i mean to ask that if i took my ethernet module outside the chip and then interfaced it again to the leon3 processor..will it work

Ah, got it! You're asking about a **custom external Ethernet MAC/PHY** interfaced back to LEON3, replacing the built-in GRETH core. In that case:

Short Answer: No, this code won't work as-is — but it's fixable.

Why it breaks

The test code is hardwired to talk to **GRETH's specific register set**. If you replace GRETH with your own external Ethernet module, your module will have:

- A **different base address** (wherever you map it on the APB/AHB bus)
- A **different register layout** (your own CTRL, STATUS, TX/RX descriptor format)
- Possibly a **different bus interface** (AHB, APB, SPI, parallel — depending on how you connect it)
- A **different descriptor/DMA mechanism** (or no DMA at all — maybe you use FIFO-based transfer)

So the test code would be writing to wrong addresses and expecting register bits that don't exist in your module.

What stays the same vs what changes

Part of the code	Stays the same?	Why
<code>printf</code> / <code>check()</code> / <code>delay()</code>	✔ Yes	Pure software, CPU-independent
<code>REPORT_START</code> / <code>REPORT_END</code> macros	✔ Yes	GRLIB testbench mechanism, not GRETH-specific
<code>sparc-gaisler-elf-gcc</code> compilation	✔ Yes	Still running on LEON3
<code>make vcs-launch</code> flow	✔ Yes	Simulation flow unchanged
<code>GRETH_BASE</code> and all register addresses	✗ No	Your module lives at a different address
<code>GRETH_CTRL/STATUS/MDIO</code> register bits	✗ No	Your register map will differ
DMA descriptor structure (<code>desc_t</code>)	✗ Probably not	Depends on your module's DMA design
MDIO read/write functions	⚠ Maybe	Only if your module still uses standard MDIO for PHY access
PHY loopback via MDIO	⚠ Maybe	Works if you kept a standard MII PHY

What you'd need to rewrite

Essentially you'd replace the **hardware abstraction layer** — the register definitions and low-level driver functions — while keeping the test logic and pass/fail structure:

```
c
// Change this block to match YOUR module's memory map
#define MY_ETH_BASE      0x80000X00    // wherever you mapped it

#define MY_ETH_CTRL      (MY_ETH_BASE + 0x00)
#define MY_ETH_STATUS    (MY_ETH_BASE + 0x04)
// ... your register layout here
```

```
// Rewrite mdio_read/write if your PHY interface changed
// Rewrite descriptor setup if your DMA format changed
```

The test structure (5 tests, check(), pass/fail summary) can stay exactly the same — you'd just be poking your registers instead of GRETH's.

The key question to ask yourself

How did you interface your external Ethernet module back to LEON3?

- If via **APB** → relatively easy remap, same coding style
- If via **AHB** → similar, but descriptor/DMA handling may change
- If via **SPI or parallel GPIO** → bigger rewrite; you'll need a software driver loop instead of DMA descriptors
- If you kept a **standard MII/RMII PHY** → the MDIO functions are reusable as-is

Want me to help you write the adapted test code once you know your module's register map and bus interface?

i won't be replacing the GRETH with my own ethernet..I will be taking the GRETH outside and convert it's apb and ahb controls into memory interface and then use a ahb/apb to memory interface bridge convertor to access the apb and ahb signals inside the check and connect them to the GRETH outside(now has memory interface) via the bridge..will this test work for that?

Ah now I understand completely! So your architecture is:

```
LEON3 SoC
├─ AHB/APB bus (internal)
├─ AHB→Memory bridge (inside chip)
└─ Memory interface pins → [off-chip] → GRETH (external, memory-mapped)
```

Short Answer: Yes, this test code will work — with no changes (ideally) or very minor ones.

Why it still works

The test code **doesn't care** where GRETH physically lives. It only does this:

```
c
WR(0x80000D00, value); // writes to an address
RD(0x80000D00);        // reads from an address
```

As long as your AHB/APB → Memory bridge correctly maps address `0x80000D00` to the external GRETH's corresponding register, LEON3 has no idea GRETH is now sitting outside the chip. The memory-mapped abstraction is transparent to the CPU. This is exactly what memory-mapped I/O bridges are designed for.

What the bridge must guarantee

For the test to pass without any code changes, your bridge must preserve:

Requirement	Why the test depends on it
Address mapping is 1:1 — <code>0x80000D00</code> still reaches GRETH CTRL	Every <code>RD()</code> / <code>WR()</code> in the test uses hardcoded addresses
Read/write semantics are preserved — no extra latency that drops data	<code>mdio_wait()</code> polls in a tight loop; stale reads will cause false timeouts
Descriptor physical addresses are still valid — <code>txd[]</code> and <code>rx[]</code> addresses that GRETH's AHB DMA uses must be reachable	Tests 4 & 5 pass <code>(unsigned int)txd</code> directly to <code>GRETH_TXDESC</code>
AHB DMA path is preserved — GRETH's DMA must still reach LEON3's SRAM	GRETH fetches frame data from <code>txbuf[]</code> in LEON3 memory over AHB

The one thing that could cause issues

The DMA path is the tricky part. Look at Test 4:

```
c
txd[0].addr = (unsigned int)txbuf[0]; // address in LEON3 SRAM
WR(GRETH_TXDESC, (unsigned int)txd); // GRETH will DMA-fetch from this address
```

GRETH (now external) needs to DMA back into LEON3's SRAM over AHB to fetch the frame data. So your bridge needs to handle both directions:

LEON3 CPU → bridge → GRETH registers (register access, APB)
GRETH DMA → bridge → LEON3 SRAM (data fetch, AHB – reverse direction!)

If your bridge only handles CPU→GRETH direction and **not** the reverse DMA path, Tests 4 & 5 will fail (TX/RX never complete) while Tests 1, 2, 3 will pass fine.

Summary

Test	Will it pass?
Test 1 — Register access	✔ Yes — pure APB reads/writes through bridge
Test 2 — MDIO read	✔ Yes — APB only
Test 3 — PHY loopback	✔ Yes — APB only
Test 4 — Single frame loopback	✔ Only if bidirectional DMA path works
Test 5 — Multi-frame stress	✔ Only if bidirectional DMA path works

So essentially this test is a **perfect validation tool for your bridge design** — if all 5 tests pass, your bridge is correctly handling both APB register access and AHB DMA in both directions. If Tests 1–3 pass but 4–5 fail, you know exactly where the bridge has a gap.

ok...can i try running this on vc code now? will it give any output or no?

No, you cannot run this on VS Code and get meaningful output. Here's why:

Why it won't work on VS Code / your PC

This is **not a regular C program**. It is **bare-metal embedded firmware** written for a specific hardware target. Three fundamental blockers:

1. The addresses don't exist on your PC

c

WR(0x80000D00, value) // writes to LEON3's APB bus address
RD(0x80000200) // GRLIB testbench report address

On your PC, address `0x80000D00` is either unmapped memory or belongs to some random OS process. Writing to it will cause an **immediate segfault / access violation** and the program will crash on the very first register access.

2. The compiler target is wrong

The code is meant to be compiled with:

```
sparc-gaisler-elf-gcc ← SPARC architecture cross-compiler
```

If you compile it with a normal PC compiler (gcc/clang/MSVC), you get an x86 or ARM binary — wrong instruction set entirely, and none of the hardware assumptions hold.

3. There is no GRETH, no PHY, no AHB bus on your PC

Even if you bypassed the address crash, there is no Ethernet MAC core, no MDIO bus, no DMA descriptor engine — the program would just be poking random memory and reading garbage back.

What you CAN do on VS Code / PC

Goal	What to do
Just check if the C code compiles without errors	Install <code>gcc</code> and compile with <code>-c</code> flag (compile only, don't link or run) — you'll catch syntax errors
Simulate the logic in software	Mock the <code>RD()</code> / <code>WR()</code> macros to use a local array instead of real addresses — useful for unit testing the logic
Actually run the test	You need VCS + GRLIB on a Linux machine with the LEON3 design

The only right way to run this

```
Linux machine
├─ GRLIB design folder (GR-XC3S-1500 template)
│   └─ make soft          ← cross-compiles with sparc-gaisler-elf-gcc
└─ make vcs-launch       ← runs VCS simulation, prints output to terminal
```

VS Code is fine as a **code editor** to read and modify `greth_test.c` — just don't try to build or run it there against real hardware. Would you like help setting up the VCS + GRLIB flow on your Linux machine?

i used make soft CC="/path/bin/sparc-gaisler-elf-gcc" XCFLAGS="-O2 -mcpu=leon3" OBJCOPY="/path/bin/sparc-gaisler-elf-objcopy" to create the srec file for the current already existing systest.c...so can i run this for my test.c too?

Yes, absolutely! The exact same command will work for your `greth_test.c` — you just need to swap the file in. Here's exactly what to do:

Steps

Step 1 — Replace systest.c

```
bash

cp greth_test.c /path/to/your/design/folder/systest.c
```

Just overwrite the existing `systest.c` with your file. The Makefile always looks for `systest.c` by that exact name.

Step 2 — Run the exact same command you already used

```
bash

make soft \
  CC="/path/bin/sparc-gaisler-elf-gcc" \
  XCFLAGS="-O2 -mcpu=leon3" \
  OBJCOPY="/path/bin/sparc-gaisler-elf-objcopy"
```

No changes needed. It will produce a new `.srec` file compiled from your `greth_test.c`.

Step 3 — Launch simulation as usual

```
bash

make vcs-launch
```

Why it works the same way

Your `greth_test.c` uses:

- Only **standard C** (`stdio.h`, `stdlib.h`, `string.h`) — all available in the SPARC GCC toolchain

- Only **memory-mapped register access** — no OS calls, no Linux-specific code
- The **same GRLIB conventions** (`REPORT_START` / `REPORT_END` , hardcoded APB addresses)

The Makefile doesn't care what's inside `systest.c` — it just cross-compiles whatever file has that name.

One thing to double-check

Make sure your `greth_test.c` has the correct **GRETH base address** matching your design's `config.vhd`:

```
c
#define GRETH_BASE 0x80000D00 // ← confirm this matches your APB slot
```

If your design maps GRETH to a different APB slave slot, this is the only line you'd need to change before running `make soft`. You can verify it by checking:

```
bash
grep -i greth /path/to/design/config.vhd
```

Everything else will carry over directly from your existing working flow.

i ran this commnand in the folder that held my test.c today..but it was not inside the srec file and i did not change the systest.c..it worked...HOW? that folder had no makefile

Great observation! Here's what most likely happened:

The Makefile was found in a Parent Directory

`make` doesn't just look in the current folder. When you run `make soft` in a folder with **no Makefile**, it automatically **walks up the directory tree** looking for one:

```
/home/user/grlib/designs/grxc3s1500/greth_test.c ← you were here
/home/user/grlib/designs/grxc3s1500/             ← no Makefile here
/home/user/grlib/designs/                         ← no Makefile here
/home/user/grlib/                                 ← FOUND Makefile here ✓
```

So `make` found and used the **GRLIB top-level Makefile** from a parent folder — and that Makefile already knew where `systest.c` lives and compiled it instead of your `greth_test.c`.

So what actually got compiled?

Almost certainly the **original existing** `systest.c` from wherever the Makefile expected it — not your `greth_test.c`. Your file just happened to sit in the same folder and was ignored.

You can confirm this by checking:

```
bash

# See which Makefile was actually used
make soft --print-directory

# Check the timestamp on the srec file – did it actually rebuild?
ls -lt *.srec

# Check what source file the Makefile points to
grep -i "systest" /path/to/found/Makefile
```

The right way to make sure YOUR file gets compiled

```
bash

# Step 1 – find where systest.c actually lives
find /path/to/grlib -name "systest.c"

# Step 2 – replace it with your file
cp greth_test.c /found/path/systest.c

# Step 3 – run make from the correct design folder
cd /correct/design/folder
make soft CC="/path/bin/sparc-gaisler-elf-gcc" \
          XCFLAGS="-O2 -mcpu=leon3" \
          OBJCOPY="/path/bin/sparc-gaisler-elf-objcopy"
```

Bottom line

What you thought happened

What actually happened

Your `greth_test.c` was compiled

The existing `systest.c` was compiled

The Makefile was in your folder

`make` found a parent directory Makefile

A new srec was generated for your test

The old srec was likely reused or rebuilt from `systest.c`

The srec that got generated **does not contain your test code** — it contains whatever was already in `systest.c`. To verify, check if the srec file's timestamp changed when you ran the command, or just do the `find` + `cp` steps above to be sure.

the folder where my file was inside the downloads..how was it able to access the Makefile? Also it seemed like the values were different only?....

Ah, if your file was in **Downloads** (`/home/user/Downloads/greth_test.c`), then the parent directory walk theory is almost certainly **wrong** — there would be no GRLIB Makefile anywhere up that path.

Most Likely What Actually Happened

You probably **already had a terminal open in the correct GRLIB design folder** from earlier work, and when you ran the command, it executed **in that terminal's current directory** — not in Downloads.

Terminal 1 (still open from before)

└─ current directory: `/home/user/grlib/designs/grxc3s1500/` ← make ran HERE
not in

Downloads

The file in Downloads was just sitting there untouched.

About the "values were different"

That's the more important clue. If the output values looked different from what the original `systest.c` would produce, there are a few possibilities:

Possibility 1 — `systest.c` was already your file

Maybe you had **already copied it** to the design folder in a previous session and forgot — so `make` did compile your code, just from the design folder's copy of `systest.c`, not from Downloads.

Possibility 2 — The existing `systest.c` was just a different test

The original `systest.c` in the design folder may simply have been a **different version or different test** — different register values, different checks — so output naturally looked different.

How to confirm exactly what happened

```
bash

# 1. Find ALL systest.c files on your machine
find / -name "systest.c" 2>/dev/null

# 2. Find where the srec was actually generated
find / -name "*.srec" -newer ~/Downloads/greth_test.c 2>/dev/null

# 3. Check which directory make actually ran in
# Look at your terminal history
history | grep "make soft"
```

The safest way to be 100% sure your file is what gets compiled

```
bash

# 1. Find the real design folder
find / -name "Makefile" 2>/dev/null | xargs grep -l "systest" 2>/dev/null

# 2. Copy your file there as systest.c
cp ~/Downloads/greth_test.c /found/design/folder/systest.c

# 3. cd into that folder and run make
cd /found/design/folder
make soft CC="/path/bin/sparc-gaisler-elf-gcc" \
          XCFLAGS="-O2 -mcpu=leon3" \
          OBJCOPY="/path/bin/sparc-gaisler-elf-objcopy"

# 4. Confirm the srec was freshly built
ls -lt *.srec
```

This way there is **zero ambiguity** about which file got compiled.

instead of copying my code into the systest.c...is there any way i can make the Makefile run my code?

Yes! There are a few ways to do this without touching `systest.c`:

Option 1 — Override the source file variable (easiest, no Makefile editing)

Most GRLIB Makefiles use a variable for the source file. Try:

```
bash
make soft SYSTEST=greth_test \
    CC="/path/bin/sparc-gaisler-elf-gcc" \
    XCFLAGS="-O2 -mcpu=leon3" \
    OBJCOPY="/path/bin/sparc-gaisler-elf-objcopy"
```

or

```
bash
make soft SRC=greth_test.c \
    CC="/path/bin/sparc-gaisler-elf-gcc" \
    XCFLAGS="-O2 -mcpu=leon3" \
    OBJCOPY="/path/bin/sparc-gaisler-elf-objcopy"
```

This works **without editing anything** if the Makefile uses that variable name internally.

Option 2 — Check what variable name the Makefile actually uses

```
bash
grep -i "systest\|SRC\|SOURCE" /path/to/Makefile
```

This will show you the exact variable name, for example:

```
makefile
SYSTEST = systest      < then use SYSTEST=greth_test
# or
```

```
SRCS      = systest.c      ← then use SRCS=greth_test.c
```

Then use that exact variable name in your `make` command.

Option 3 — Create a symlink (clean, no file copying)

```
bash

cd /path/to/design/folder

# rename original systest.c temporarily
mv systest.c systest_original.c

# create a symlink pointing to your file
ln -s ~/Downloads/greth_test.c systest.c

# now run make normally
make soft CC="/path/bin/sparc-gaisler-elf-gcc" \
          XCFLAGS="-O2 -mcpu=leon3" \
          OBJCOPY="/path/bin/sparc-gaisler-elf-objcopy"

# restore original when done
rm systest.c
mv systest_original.c systest.c
```

The Makefile sees `systest.c` as normal but it actually points to your file in Downloads. **Nothing is overwritten.**

Option 4 — Edit the Makefile directly (permanent change)

If you want a permanent fix, open the Makefile and find the line that references `systest`:

```
makefile

SYSTEST = systest
```

Change it to:

```
makefile

SYSTEST = greth_test
```

Option	Best when
Option 1	Try first — zero effort
Option 2	Option 1 didn't work — need to find the right variable name
Option 3	You want zero permanent changes to anything
Option 4	This is your dedicated test folder and you want it permanent

Try **Option 1 first** — if the Makefile uses a standard GRLIB variable it'll just work.

ok i will try option 1 ..Also once i do that and i launch the make vcs-launch...how should i make it visible in the waveform ? how should i check it's working...should i change the ram.srec parts in all files in the codes to my new srec?

Great questions — let me break this into two parts:

Part 1 — The SREC file

Yes, if your design files reference `ram.srec` by name, you need to make sure they point to the **newly generated srec from your greth_test compile**.

First check what your srec is named

When you ran `make soft` with Option 1, the output srec will likely be named after your file:

```
greth_test.srec    ← if SYSTEST=greth_test was used
# or still
systest.srec      ← if Makefile hardcodes the output name
```

Check with:

```
bash
ls -lt *.srec      # newest file at top
```

Where ram.srec is referenced


```
bash

# Find all files that reference ram.srec
grep -r "ram.srec" /path/to/design/folder/
```

Typically it appears in files like:

```
testbench.vhd      ← VHDL testbench memory loader
vsim.do / vcs.do   ← simulator script
Makefile           ← build script
modelsim.ini       ← sometimes here too
```

What to do

Option A — Rename your srec to ram.srec (easiest)

```
bash

cp greth_test.srec ram.srec
```

Option B — Change the reference in the do/tcl script

```
bash

# find the line and change it
grep -n "ram.srec" vcs.do
# then edit that line to point to greth_test.srec
```

Part 2 — Seeing it in the Waveform

This is the most important part. Here's exactly what to do:

Step 1 — Add the right signals to the waveform

When `make vcs-launch` opens the VCS DVE waveform viewer, you need to add these signal groups:

GRETH APB Registers — to see register reads/writes

```
# In DVE, browse to:
testbench → leon3soc → apbctrl → greth0
    → psel
    → penable
    → pwrite
    → paddr      ← which register is being accessed
```

→ pwdata ← data being written

→ prdata ← data being read back

GRETH AHB DMA — to see frame data moving

testbench → leon3soc → greth0

→ ahbmi ← AHB master input (DMA reads frame data)

→ ahbmo ← AHB master output

GRETH TX/RX — to see actual Ethernet activity

testbench → leon3soc → greth0

→ txd ← transmit data lines

→ tx_en ← transmit enable

→ rxd ← receive data lines

→ rx_dv ← receive data valid

MDIO — to see PHY communication (Tests 2 & 3)

testbench → leon3soc → greth0

→ mdc ← MDIO clock

→ mdio_o ← MDIO data out

→ mdio_i ← MDIO data in

Step 2 — What each test looks like on the waveform

Test	What to look for
Test 1 — Register access	psel goes high, pwrite toggles, paddr shows 0x00, 0x08, 0x0C — MAC address registers being written
Test 2 — MDIO read	mdc clock toggles, mdio_o sends read command bits serially, mdio_i returns PHY ID bits
Test 3 — PHY loopback	mdio_o sends write command with loopback bit set (0x6100)
Test 4 — Single frame	tx_en pulses high for 64 bytes, then rx_dv pulses high as loopback frame arrives
Test 5 — Multi frame	tx_en pulses 4 times back to back, rx_dv responds 4 times

Step 3 — What to check for PASS confirmation

On the waveform look for these healthy signs:

- ✓ psel/penable toggling cleanly → APB bus working through your bridge
- ✓ mdc clock running → MDIO active
- ✓ tx_en going high → GRETH actually transmitting
- ✓ rx_dv going high after tx_en → loopback working
- ✓ No X (unknown) values on data bus → no connectivity issues
- ✓ TXIRQ/RXIRQ status bits going high → DMA completing

And the **red flags** to watch for:

- ✗ X values on paddr/prdata correctly → bridge not propagating signals
- ✗ tx_en never goes high → DMA path broken (Tests 4/5 issue)
- ✗ rx_dv never responds connected → PHY loopback not set or MII not
- ✗ mdc never toggles → MDIO bus issue

Step 4 — Saving the waveform setup

Once you've added all the signals, save the setup so you don't redo it every time:

```
tcl

# In DVE console
save_session -file greth_waves.tcl

# Next time load it with
make vcs-launch
# then in DVE: File → Restore Session → greth_waves.tcl
```

Quick summary of the flow

```
make soft SYSTEST=greth_test ... ← compiles your code → greth_test.srec
cp greth_test.srec ram.srec      ← make srec available to testbench
make vcs-launch                  ← runs simulation
DVE opens → add signals → watch waveform
```

The **transcript output** (PASS/FAIL printed lines) tells you if the test logic passed, and the **waveform** tells you exactly *why* — which signals moved, which didn't, and where

your bridge may have a gap. Both together give you the full picture.

explain the tests in a simple manner

Sure! Let me explain all 5 tests in plain simple terms:

Think of it like this first

Imagine GRETH is a **post office** inside your chip. It sends and receives letters (Ethernet frames). The tests basically check:

- Is the post office building there? (registers)
 - Can you talk to the postman? (MDIO/PHY)
 - Can you actually send and receive a letter? (frames)
-

Test 1 — Is GRETH alive and can we configure it?

What it does in plain English:

- Restarts GRETH (like switching it off and on)
- Writes the chip's MAC address (like writing your home address on the post office door)
- Turns on TX and RX (opens the send and receive windows)
- Reads everything back to confirm it was saved correctly

Pass means: GRETH is physically accessible and its registers work correctly through your bus/bridge.

Test 2 — Can GRETH talk to the PHY chip?

What the PHY is: The PHY is the actual physical Ethernet chip that converts digital signals to electrical signals on the wire. GRETH talks to it over a 2-wire serial bus called **MDIO** — think of it like GRETH calling the PHY on a phone.

What it does in plain English:

- Asks the PHY "what is your ID number?" over MDIO
- Checks the PHY's status
- Confirms the phone line (MDIO bus) is working

Pass means: GRETH and PHY can communicate — the MDIO bus is healthy.

Test 3 — Tell the PHY to echo everything back

What it does in plain English:

- Sends a command to the PHY over MDIO saying "**whatever I send you, send it straight back to me**"
- This is called **loopback mode** — like talking into a walkie talkie that immediately plays your voice back
- Reads back the PHY to confirm it actually accepted the loopback command

Pass means: PHY is now in loopback mode — ready for the actual frame tests below.

Test 4 — Send one letter and get it back

What it does in plain English:

- Writes one 64-byte Ethernet frame into memory (like writing a letter)
- Tells GRETH "here is the letter, send it"
- GRETH sends it to the PHY → PHY echoes it straight back (because of Test 3) → GRETH receives it
- Opens the received letter and checks **every single byte** matches what was sent

Pass means: The full send → loopback → receive → verify path works end to end.

Test 5 — Send 4 letters back to back

What it does in plain English:

- Does exactly what Test 4 did but with **4 different frames at the same time**
- Each frame has a **different payload pattern** so you can tell them apart
- Waits for all 4 to be sent and all 4 to come back
- Verifies all 4 received frames match their originals exactly

Pass means: GRETH can handle multiple frames in a row without mixing up data or dropping anything — this is the **stress test**.

One line summary of each

Test	One line
Test 1	Is GRETH reachable and configurable?
Test 2	Can GRETH talk to the PHY over MDIO?
Test 3	Put PHY into echo/loopback mode
Test 4	Send 1 frame, get it back, check it
Test 5	Send 4 frames, get them all back, check all of them

Each test **builds on the previous one** — if Test 2 fails, Tests 3/4/5 will also fail. So always fix failures from the top down.

can i tell me what u are sending ? like the values i can expect to see

Sure! Let me break down exactly what bytes are being sent on the wire.

The Ethernet Frame Structure

Every frame sent in this test looks like this:

[DST MAC 6 bytes][SRC MAC 6 bytes][EtherType 2 bytes][Payload 50 bytes]

Total = 64 bytes (minimum legal Ethernet frame size)

Fixed Header (same in ALL frames, Tests 4 & 5)

Destination MAC — bytes 0 to 5

FF:FF:FF:FF:FF:FF

This is the **broadcast address** — sent to everyone on the network. Used here so the PHY/GRETH accepts the frame back in promiscuous mode.

Source MAC — bytes 6 to 11

02:00:00:00:00:08

This is a **locally administered MAC address** — the same one written to GRETH in Test 1.

EtherType — bytes 12 to 13

08:00

This means **IPv4** — not because we're actually sending IP, just a valid EtherType placeholder.

The Payload — bytes 14 to 63

This is where frames **differ between Test 4 and Test 5**.

The formula in the code is:

```
c
buf[j] = (unsigned char)(frame_id * 0x10 + j);
```

Where `frame_id` is 0 for Test 4, and 0/1/2/3 for Test 5's four frames.

Test 4 — Single frame (frame_id = 0)

```
frame_id = 0
formula  = (0 × 0x10) + byte_position = just the byte position
```

Byte position	Value (hex)	Value (decimal)
14	0x0E	14
15	0x0F	15
16	0x10	16
17	0x11	17
18	0x12	18
...	...	...
63	0x3F	63

So the payload is simply **counting upward from 0x0E to 0x3F**.

Test 5 — 4 frames (frame_id = 0, 1, 2, 3)

Each frame has a different **starting offset** because of `frame_id × 0x10`:

Frame 0 (frame_id = 0) — same as Test 4:

```
byte 14 = 0x0E
byte 15 = 0x0F
byte 16 = 0x10
...
byte 63 = 0x3F
```

Frame 1 (frame_id = 1):

```
formula = (1 × 0x10) + position = 0x10 + position

byte 14 = 0x10 + 14 = 0x1E
byte 15 = 0x10 + 15 = 0x1F
byte 16 = 0x10 + 16 = 0x20
...
byte 63 = 0x10 + 63 = 0x4F
```

Frame 2 (frame_id = 2):

```
formula = (2 × 0x10) + position = 0x20 + position

byte 14 = 0x20 + 14 = 0x2E
byte 15 = 0x20 + 15 = 0x2F
byte 16 = 0x20 + 16 = 0x30
...
byte 63 = 0x20 + 63 = 0x5F
```

Frame 3 (frame_id = 3):

```
formula = (3 × 0x10) + position = 0x30 + position

byte 14 = 0x30 + 14 = 0x3E
byte 15 = 0x30 + 15 = 0x3F
byte 16 = 0x30 + 16 = 0x40
...
byte 63 = 0x30 + 63 = 0x6F
```


Full Frame — What you will see on the waveform (Test 4)

Every single byte in order on the MII TX data pins:

FF FF FF FF FF FF	← Destination MAC (broadcast)
02 00 00 00 00 08	← Source MAC
08 00	← EtherType (IPv4)
0E 0F 10 11 12 13	← Payload start
14 15 16 17 18 19	
1A 1B 1C 1D 1E 1F	
20 21 22 23 24 25	
26 27 28 29 2A 2B	
2C 2D 2E 2F 30 31	
32 33 34 35 36 37	
38 39 3A 3B 3C 3D	
3E 3F	← Payload end (byte 63)

What to match on the waveform

When you open DVE and look at the **TXD pins**, you should see these values clocking out in nibbles (MII sends 4 bits at a time):

MII nibbles for first byte FF:	
TXD = 1111 then 1111	← 0xF then 0xF = 0xFF ✓
MII nibbles for source MAC first byte 02:	
TXD = 0000 then 0010	← 0x0 then 0x2 = 0x02 ✓
MII nibbles for EtherType 08 00:	
TXD = 0000 then 1000	← 0x08
TXD = 0000 then 0000	← 0x00 ✓
MII nibbles for first payload byte 0x0E:	
TXD = 0000 then 1110	← 0x0 then 0xE = 0x0E ✓

Quick reference card

What	Value
Destination MAC	FF:FF:FF:FF:FF:FF

What	Value
Source MAC	02:00:00:00:00:08
EtherType	0x0800 (IPv4)
Frame size	64 bytes
Payload start byte	0x0E (Test 4 / Frame 0)
Payload end byte	0x3F (Test 4 / Frame 0)
Frame 1 payload start	0x1E
Frame 2 payload start	0x2E
Frame 3 payload start	0x3E

So when you're looking at the waveform, if you see `FF FF FF FF FF FF 02 00 00 00 00 08 08 00` at the start of a transmission — **that's your frame going out correctly.**

in the code where have u assigned this

Good question! Let me point you to the exact lines in the code:

The function that builds the frame

Everything is assigned inside `fill_eth_header()` — lines 215 to 229:

```
c
static void fill_eth_header(unsigned char *buf, int frame_id)
{
    /* Destination: broadcast FF:FF:FF:FF:FF:FF */
    buf[0]=0xFF; buf[1]=0xFF; buf[2]=0xFF;
    buf[3]=0xFF; buf[4]=0xFF; buf[5]=0xFF;

    /* Source: 02:00:00:00:00:08 */
    buf[6]=0x02; buf[7]=0x00; buf[8]=0x00;
    buf[9]=0x00; buf[10]=0x00; buf[11]=0x08;

    /* EtherType: 0x0800 = IPv4 */
    buf[12]=0x08; buf[13]=0x00;
```

```
/* Payload: known pattern per frame */
int j;
for (j = 14; j < FRAME_SIZE; j++)
    buf[j] = (unsigned char)(frame_id * 0x10 + j);
}
```

Where it gets called

In Test 4 — single frame:

```
c
fill_eth_header(txbuf[0], 0); // frame_id = 0
```

`txbuf[0]` is the buffer, `0` is the `frame_id` → payload = `0x10*0 + j` = just `j`

In Test 5 — 4 frames:

```
c
for (i = 0; i < NUM_FRAMES; i++) {
    fill_eth_header(txbuf[i], i); // frame_id = 0,1,2,3
}
```

Each loop pass `i` becomes the `frame_id` → different payload per frame

Where the buffers are declared

At the top of the file:

```
c
// frame data buffers — 32 byte aligned for AHB
static unsigned char txbuf[NUM_FRAMES][128]; // TX buffers
static unsigned char rxbuf[NUM_FRAMES][128]; // RX buffers
```

So `txbuf` is a **4×128 array** — 4 frames, 128 bytes each. Only the first 64 bytes (`FRAME_SIZE`) of each are actually used and sent.

Summary of which line does what

What

Where in code

<code>FF:FF:FF:FF:FF:FF</code> destination	<code>buf[0]</code> to <code>buf[5]</code>
<code>02:00:00:00:00:08</code> source MAC	<code>buf[6]</code> to <code>buf[11]</code>
<code>0x0800</code> EtherType	<code>buf[12]</code> and <code>buf[13]</code>
Payload pattern	<code>for loop</code> — <code>buf[14]</code> to <code>buf[63]</code>
Test 4 calls it	<code>fill_eth_header(txbuf[0], 0)</code>
Test 5 calls it	<code>fill_eth_header(txbuf[i], i)</code> inside for loop

in test 1 2 3 no values in txd and rxd?

Exactly right! Tests 1, 2, and 3 have **zero activity on TXD and RXD pins**. Here's why:

What each test actually uses

Test	APB registers	MDIO bus	TXD/RXD pins	DMA/Descriptors
Test 1	✔ Yes	✗ No	✗ No	✗ No
Test 2	✔ Yes	✔ Yes	✗ No	✗ No
Test 3	✔ Yes	✔ Yes	✗ No	✗ No
Test 4	✔ Yes	✗ No	✔ Yes	✔ Yes
Test 5	✔ Yes	✗ No	✔ Yes	✔ Yes

Why Tests 1, 2, 3 don't touch TXD/RXD

Test 1

Only writes and reads **APB registers** — CTRL, MACMSB, MACLSB. No frame is built, no descriptor is set up, no transmission is triggered. It's purely:

c

```
WR(GRETH_CTRL, ...)    // APB write
RD(GRETH_MACMSB)       // APB read
```

Nothing reaches the MII pins at all.

Test 2

Only uses the **MDIO serial bus** — a completely separate 2-wire interface (MDC clock + MDIO data) used just for PHY configuration. It has nothing to do with the actual Ethernet data path:

```
c
mdio_read(PHY_ADDR, PHY_ID1)  // goes out on MDC/MDIO pins only
```

TXD stays flat/zero the entire time.

Test 3

Same as Test 2 — only MDIO activity:

```
c
mdio_write(PHY_ADDR, PHY_CTRL, PHY_CTRL_LOOPBACK | ...) // MDIO only
```

Still nothing on TXD/RXD.

So on your waveform for Tests 1, 2, 3:

TXD	_____	(flat, no activity)
TX_EN	_____	(always 0)
RXD	_____	(flat, no activity)
RX_DV	_____	(always 0)
MDC	┌─┐┌─┐┌─┐┌─┐┌─┐	(toggling only during Tests 2 & 3)
MDIO	┌─┐┌─┐┌─┐┌─┐┌─┐	(serial PHY data only during Tests 2 & 3)
PSEL	┌─┐┌─┐┌─┐┌─┐┌─┐	(APB select – active all through Tests 1, 2, 3)
PADDR	(shows 0x00, 0x08, 0x0C, 0x10 – register addresses)	

The simple way to think about it

Bus/Signal	What it carries	Active in
APB (PSEL/PADDR/PWDATA)	Register config commands	Tests 1, 2, 3, 4, 5
MDIO (MDC/MDIO)	PHY configuration	Tests 2, 3 only
MII (TXD/RXD)	Actual Ethernet frame data	Tests 4, 5 only
AHB DMA	Frame data moving in memory	Tests 4, 5 only

Tests 1-3 are purely **configuration and setup**. The Ethernet data pins only wake up in **Test 4** when the very first frame is actually transmitted.

what errors can i expect

Great question! Here are all the errors you can expect, grouped by where they come from:

Category 1 — Errors Printed by the Test Code Itself

These appear directly in the VCS transcript as FAIL lines:

Test 1 Failures

FAIL: GRETH reset bit self-clears

Means: Reset bit never cleared — GRETH is not responding at all. Either wrong base address or bridge not working.

FAIL: MAC MSB write-readback
FAIL: MAC LSB write-readback

Means: Wrote MAC address but read back wrong value. APB write is going through but read is returning garbage — bridge may have a read path issue.

FAIL: TX enable bit set
FAIL: RX enable bit set
FAIL: Full-duplex bit set
FAIL: 100Mbit bit set

Means: CTRL register bits not sticking — GRETH CTRL register not accessible properly.

Test 2 Failures

FAIL: PHY ID1 not 0xFFFF (MDIO responding)
FAIL: PHY ID2 not 0xFFFF (MDIO responding)

Means: MDIO read returned 0xFFFF — PHY is not responding. MDIO bus not connected or PHY model not instantiated in testbench.

FAIL: PHY STATUS readable

Means: mdio_read() timed out and returned 0xDEAD — MDIO busy bit never cleared, meaning MDIO transaction never completed.

FAIL: No MDIO link failure

Means: GRETH detected a link failure on MDIO — MDC/MDIO lines may be floating or PHY not properly connected.

Test 3 Failures

FAIL: PHY loopback bit confirmed via readback

Means: Wrote loopback bit to PHY CTRL but readback shows it didn't stick. PHY model may not support loopback control or MDIO write path is broken.

Test 4 Failures

FAIL: Single frame TX completed

Means: Polled TXIRQ for 500000 cycles and it never came. Most common cause — DMA path broken, descriptor not set up correctly, or GRETH TX not enabled properly.

FAIL: No TX errors

Means: STS_TXERR or STS_TXAHB bit set — GRETH tried to transmit but got an AHB bus error fetching frame data from LEON3 memory. Reverse DMA path

(GRETH→LEON3 SRAM) not working.

FAIL: Single frame RX completed

Means: TX worked but frame never came back. PHY loopback not active (Test 3 failed silently) or RX descriptor not set up correctly.

FAIL: No RX errors

Means: STS_RXERR or STS_RXAHB set — GRETH received a frame but got AHB error writing it to rxbuf in LEON3 memory.

FAIL: Loopback frame payload matches
Data mismatch at byte N (TX=0xXX RX=0xYY)

Means: Frame came back but data is corrupted. Tells you exactly which byte is wrong — very useful for debugging bridge data integrity.

Test 5 Failures

FAIL: All TX frames sent
FAIL: No TX errors (stress)
FAIL: All RX frames received
FAIL: No RX errors (stress)
FAIL: All frame payloads match (stress)
Frame N: byte M mismatch (TX=0xXX RX=0xYY)
Frame N still owned by HW

Means: Same as Test 4 but across multiple frames. "Still owned by HW" means DESC_EN bit never cleared — GRETH never processed that descriptor.

Category 2 — Simulation Level Errors (in VCS transcript)

These come from the VHDL testbench, not your C code:

ERROR: AHB error response

Means: Something on the AHB bus returned an error response — usually your bridge returning HRESP=ERROR when it shouldn't.

ERROR: Timeout waiting for interrupt

Means: Testbench level timeout — simulation ran too long without the LEON3 finishing.

WARNING: X propagation on GRETH signals

Means: Unknown (X) values reaching GRETH inputs — uninitialized signals in your bridge or memory interface.

ERROR: REPORT address written with unexpected value

Means: The testbench saw an unexpected value at 0x80000200 — test finished in an unexpected state.

Category 3 — Waveform Level Problems

These won't print as errors but you'll see them on the waveform:

What you see	What it means
X values on TXD/RXD	Uninitialized or unconnected signals in bridge
TXD toggling but RXD always flat	PHY loopback not working
PSEL never goes high	Bridge not decoding GRETH address correctly
MDC never toggles	MDIO clock not reaching PHY through bridge
TX_EN pulses but RX_DV never comes	MII loopback path broken
Correct TX but garbled RX data	Bridge introducing bit errors on return path
AHB signals stuck	Bridge deadlocked — may need reset

Category 4 — Compile / Setup Errors (before simulation even starts)

sparc-gaisler-elf-gcc: command not found

Toolchain not in PATH.

undefined reference to `main'

Wrong linker script — GRLIB startup code not included.

```
srec file not found
```

Testbench looking for `ram.srec` but your file has a different name.

```
VCS compile error: cannot find greth_test.c
```

Makefile SYSTEST variable not matching your filename.

Most Likely Errors for YOUR Specific Setup

Since you have an **external GRETH connected via a memory bridge**, the most probable failures are:

Priority	Error	Why likely for you
1st	<code>FAIL: GRETH reset bit self-clears</code>	Bridge address mapping not correct
2nd	<code>FAIL: Single frame TX completed</code>	Reverse DMA path (GRETH→LEON3) not working
3rd	<code>FAIL: Loopback frame payload matches</code>	Bridge introducing data corruption
4th	<code>X values on waveform</code>	Bridge output signals not initialized
5th	<code>FAIL: No TX errors</code>	AHB error response from bridge

Fix them **top to bottom** — if Test 1 fails, don't even worry about Tests 4 and 5 yet. Each test builds on the previous one.

(TEST 1) FAIL : 100Mbit set , (TEST 2) FAIL: No MDIO link failure , (TEST 4) FAIL: Loopback frame payload matches , (In TEST 5) No Tx errors , FAIL: All Rx frames received, FAIL: All frame payloads match...Frame 0...3 still owned by HW. I HAD 2 MORE ERROR (TEST3) PHY Loopback bit confirmed via readback (TEST 4) No Tx errors. BUT I fixed these changing the delay(10000) in Test 3 to while(CTRL_MDIO & 0X8) and it worked. Keep in mind the OG Systest.c has a GRETH test and it passes just fine

Great debugging work on Tests 3 & 4! Now let's analyze the remaining failures systematically. The key insight here is:

Original systest.c passes fine → so GRETH hardware, PHY, MII, DMA are all working correctly. The problem is in how YOUR test code is talking to them.

Failure Analysis One by One

TEST 1 — FAIL: 100Mbit bit set

This is a small but important clue. Everything else in Test 1 passed — only the 100Mbit bit failed.

What the code does:

```
c
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD | CTRL_100MB);
v = RD(GRETH_CTRL);
check("100Mbit bit set", v & CTRL_100MB); // checks bit 7
```

Most likely cause: The original systest.c probably sets up GRETH in **10Mbit mode** not 100Mbit. Check what speed the PHY simulation model is configured for:

```
bash
grep -i "100\\|speed\\|mbit" /path/to/testbench.vhd
grep -i "100\\|speed\\|mbit" /path/to/eth_phy.vhd
```

The PHY model in simulation may be **auto-negotiating to 10Mbit** and the GRETH hardware is overriding your 100Mbit bit to match. Try changing your CTRL write to:

```
c
// Remove CTRL_100MB and test at 10Mbit first
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD);
```

If that passes, your simulation PHY model is 10Mbit only.

TEST 2 — FAIL: No MDIO link failure

What the code checks:

```
c
```

```
check("No MDIO link failure", !(RD(GRETH_MDIO) & MDIO_LINKFAIL));
```

This is directly connected to the 100Mbit failure above. Here is why:

The PHY is reporting a **link failure because GRETH is trying to run at 100Mbit but the PHY negotiated 10Mbit** — speed mismatch = no link = MDIO_LINKFAIL bit gets set.

Fix the 100Mbit issue in Test 1 first and this will very likely clear itself automatically.

TEST 3 & 4 (already fixed)

Good fix replacing `delay(10000)` with `while(CTRL_MDIO & 0x8)` — you were simply not waiting long enough for the MDIO transaction to complete in simulation time. The original delay was tuned for real hardware speed, not VCS simulation speed.

Apply the same fix pattern everywhere delay is used before an MDIO operation:

```
c
// Replace all delays before MDIO operations with proper polling
// delay(10000);           ← remove this
while(RD(GRETH_MDIO) & MDIO_BUSY); // ← use this instead
```

TEST 4 — FAIL: Loopback frame payload matches

What it means: TX completed, RX completed, but received bytes don't match sent bytes — data corruption somewhere.

Most likely cause — the 100Mbit/10Mbit mismatch again. Running MII at wrong speed causes bit misalignment on the data pins, corrupting the received frame.

Fix the speed issue first. If payload still mismatches after that, then check:

```
c
// Is rxbuf being cleared properly before RX?
memset(rxbuf[0], 0, FRAME_SIZE); // this is already in the code

// Check RX descriptor buffer size
rxd[0].ctrl = DESC_EN | DESC_WRAP | 128; // 128 byte buffer — is this enough?
```

Also verify on the waveform — do the TXD values match what we calculated earlier (`FF FF FF FF FF FF 02 00...`)? If TXD looks correct but RXD is shifted or garbled, it's a speed/timing issue.

TEST 5 — FAIL: No TX errors + All RX frames received + All payloads match + Frames 0-3 still owned by HW

"Still owned by HW" is the most telling error — it means `DESC_EN` bit was **never cleared** by GRETH, meaning GRETH never processed those descriptors at all.

This is almost certainly caused by the **same 100Mbit problem cascading** — but there may also be a descriptor ring issue. Look at this code:

```
c
for (i = 0; i < NUM_FRAMES; i++) {
    txd[i].ctrl = DESC_EN | FRAME_SIZE;           // no WRAP on 0,1,2
    rxd[i].ctrl = DESC_EN | 128;

    if (i == NUM_FRAMES - 1) {
        txd[i].ctrl |= DESC_WRAP;                 // WRAP only on last
        rxd[i].ctrl |= DESC_WRAP;
    }
}
```

Potential issue: GRETH processes descriptor 0, then looks for descriptor 1 — but by the time it gets there, has the speed/timing issue caused it to lose sync? Check on waveform whether TX_EN pulses **4 times** or just **1 time**.

The Root Cause Chain

Almost all your failures trace back to **one root cause**:

```
100Mbit bit not sticking
    ↓
PHY and GRETH speed mismatch
    ↓
MDIO link failure flag set
    ↓
MII data pins running at wrong timing
    ↓
Frame payload corrupted on loopback
    ↓
Test 5 descriptors never processed correctly
```

Fix Plan — Do This in Order

Step 1 — Fix the speed mismatch first

Check what speed the original systest.c uses:

```
bash

grep -i "CTRL_100\|100MB\|speed" /path/to/original/systest.c
```

Match whatever speed it uses. Most likely:

```
c

// Change this in your test
#define CTRL_100MB (1 << 7)

// If original uses 10Mbit, remove CTRL_100MB from all WR(GRETH_CTRL...) calls
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD); // no CTRL_100MB
```

Step 2 — Apply your MDIO fix everywhere

```
c

// Anywhere you see delay() before or after MDIO
// Replace with proper busy polling
while(RD(GRETH_MDIO) & MDIO_BUSY);
```

Step 3 — Rerun and check

After fixing speed, expected results:

```
TEST 1: All 7 checks PASS
TEST 2: All 4 checks PASS (link failure clears)
TEST 3: Already fixed ✓
TEST 4: TX done, RX done, payload matches ✓
TEST 5: All descriptors processed, all payloads match ✓
```

Step 4 — If Test 5 descriptors still owned by HW

Add a small MDIO poll wait between Test 4 and Test 5 to let GRETH fully settle:

```
c

// At start of test_multi_frame_loopback()
while(RD(GRETH_MDIO) & MDIO_BUSY); // wait for any pending MDIO
WR(GRETH_CTRL, 0);                  // briefly disable
delay(1000);
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD); // re-enable
```

Quick Reference — Your Errors vs Root Cause

Your Error	Root Cause	Fix
FAIL: 100Mbit bit set	PHY model is 10Mbit only	Remove CTRL_100MB
FAIL: No MDIO link failure	Speed mismatch → no link	Fix speed first
FAIL: Loopback payload matches	MII timing wrong due to speed	Fix speed first
FAIL: No TX errors (Test 5)	Cascades from speed issue	Fix speed first
FAIL: All RX frames received	Descriptors never processed	Fix speed + check WRAP bit
Frames 0-3 still owned by HW	GRETH never cleared DESC_EN	Fix speed, verify descriptor ring

Fix the **100Mbit issue first** — there is a very high chance it resolves 80% of your remaining failures in one go.

RX is not receviing at all..it's all 00 only...thats why error i guess

Yes! That confirms it exactly. If RX buffer is all 00 it means **the frame never arrived in the RX buffer at all** — GRETH either never received it or received it but couldn't write it to memory.

Why RX is all zeros

There are 3 possible reasons:

- Reason 1: Frame never came back from PHY (loopback not working)
- Reason 2: GRETH received it but failed to DMA-write it to rxbuf (AHB path broken)
- Reason 3: RX descriptor not set up correctly (GRETH doesn't know where to put it)

How to find which one it is

Check 1 — Look at RX_DV on waveform

RX_DV stays 0 the whole time? → Reason 1 (frame never came back from PHY)
RX_DV pulses high? → Reason 2 or 3 (frame arrived but not written to memory)

Check 2 — Look at RXD pins

RXD all flat/zero? → Frame never reached GRETH physically
RXD has data on it? → Frame reached GRETH but DMA write failed

Most Likely Culprit — The 100Mbit Issue

Since the original systest.c passes, and your test fails RX completely, look at this sequence in your code:

```
c
// Test 4 enables GRETH at 100Mbit
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD | CTRL_100MB | CTRL_PROM);
```

But if PHY is in **10Mbit mode**, the MII timing is completely different:

100Mbit MII → data changes every 40ns (25MHz clock)
10Mbit MII → data changes every 400ns (2.5MHz clock)

GRETH is sampling at 100Mbit speed but PHY is sending at 10Mbit speed → **GRETH misses every single RX bit** → rxbuf stays all zeros.

Quick Confirmation Test

Open the waveform and zoom into the moment **just after TX_EN goes low** (frame was sent). Look at these 3 signals together:

TX_EN  ← frame was sent here

RX_DV  ← should pulse here if loopback working
if flat = PHY loopback not echoing back

RXD _____ ← should have data if RX_DV pulsed

The Fix

Fix 1 — Remove 100Mbit, run at 10Mbit (most likely fix)

Change every `WR(GRETH_CTRL...)` in Tests 4 and 5:

```
c
// BEFORE (your current code)
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD | CTRL_100MB | CTRL_PROM);

// AFTER (remove CTRL_100MB)
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD | CTRL_PROM);
```

Also remove from Test 1:

```
c
// BEFORE
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD | CTRL_100MB);

// AFTER
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD);
```

And the PHY loopback write in Test 3:

```
c
// BEFORE
mdio_write(PHY_ADDR, PHY_CTRL, PHY_CTRL_LOOPBACK | PHY_CTRL_100MB | PHY_CTRL_FUL

// AFTER
mdio_write(PHY_ADDR, PHY_CTRL, PHY_CTRL_LOOPBACK | PHY_CTRL_FULLD);
```

Fix 2 — Double check RX descriptor setup

Even at correct speed, if RX descriptor is wrong GRETH won't write to rxbuf. Verify:

```
c
```

```
// This must be set BEFORE enabling GRETH CTRL
rxdbuf[0].addr = (unsigned int)rxbuf[0];    // points to actual buffer
rxdbuf[0].ctrl = DESC_EN | DESC_WRAP | 128; // HW owns it, 128 byte buffer

WR(GRETH_RXDESC, (unsigned int)rxdbuf);    // tell GRETH where descriptor is

// THEN enable GRETH
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD | CTRL_PROM);
```

Order matters — if you enable CTRL before setting RXDESC, GRETH might miss the incoming frame entirely.

Fix 3 — Add a wait between TX done and checking RX

```
c

// After clearing TXIRQ
WR(GRETH_STATUS, STS_TXIRQ);

// Add this — give PHY time to loop frame back
int settle;
for(settle = 0; settle < 10000; settle++);

// Then poll for RX
for (i = 0; i < TIMEOUT; i++) {
    sts = RD(GRETH_STATUS);
    if (sts & STS_RXIRQ) { rx_ok = 1; break; }
    if (sts & STS_ERRORS) { break; }
}
```

Do This Right Now

Step 1 — Check waveform for RX_DV after TX_EN goes low:

RX_DV flat → go to Fix 1 (speed issue)
 RX_DV high but rxbuf still zero → go to Fix 2 (descriptor/DMA issue)

Step 2 — Check what speed original systest.c uses:

```
bash

grep "CTRL\|speed\|100\|10MB" /path/to/original/systest.c
```

Step 3 — Match that speed in your code and rerun.

The fact that rxbuf is **completely zero** (not partially wrong) strongly points to **Fix 1** — the frame is simply never reaching GRETH's RX path because of the speed mismatch. Fix that first before anything else.

RXD all flat/zero? → Frame never reached GRETH physically , RX_DV stays 0 the whole time? → Reason 1 (frame never came back from PHY) I think yea

Okay so the frame is **never coming back from the PHY at all**. The loopback is not working physically.

The Loopback Chain

For RX_DV to pulse, this entire chain must work:

LEON3

- writes TX descriptor
- GRETH fetches frame over AHB DMA
- GRETH drives TXD + TX_EN on MII
- PHY receives it
- PHY echoes it back (loopback mode)
- PHY drives RXD + RX_DV on MII
- GRETH receives it
- GRETH writes to rxbuf over AHB DMA

Since RXD and RX_DV are flat zero, the chain is breaking somewhere between **PHY receiving** and **PHY echoing back**.

Check TX_EN First on Waveform

Before blaming PHY loopback, confirm the frame is actually being sent out:

- TX_EN pulses? → GRETH is transmitting, PHY loopback is the problem
TX_EN flat too? → Problem is even before PHY, GRETH never transmitted

If TX_EN is also flat

GRETH never even transmitted. Most likely causes:

Cause 1 — 100Mbit speed mismatch

GRETH configured at 100Mbit but PHY clock running at 10Mbit. GRETH waits for correct clock but never gets it so TX never starts:

```
c

// Remove CTRL_100MB immediately
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD | CTRL_PROM);
```

Cause 2 — TX descriptor not seen by GRETH

GRETH_TXDESC not written before enabling CTRL:

```
c

// Make sure this order is exact
WR(GRETH_TXDESC, (unsigned int)txd); // 1st - set descriptor
WR(GRETH_RXDESC, (unsigned int)rxd); // 2nd - set descriptor
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD | CTRL_PROM); // 3rd - enable
```

Cause 3 — AHB DMA path broken

GRETH can see the descriptor but cannot fetch frame data from LEON3 SRAM. On waveform look for AHB HRESP = ERROR signal.

If TX_EN pulses but RX_DV is flat

Frame went out but PHY loopback is not echoing it back. This is the **PHY loopback mode not being set correctly**.

Check these things one by one:

Check 1 — Did Test 3 actually write the correct value?

Your fix changed the delay to a poll — but double check what value was written to PHY CTRL:

```
c

mdio_write(PHY_ADDR, PHY_CTRL,
            PHY_CTRL_LOOPBACK | PHY_CTRL_100MB | PHY_CTRL_FULLD);
```

This writes **0x6100** to PHY CTRL register. But if PHY is 10Mbit:

```
PHY_CTRL_LOOPBACK = 0x4000
PHY_CTRL_100MB    = 0x2000  ← this might be confusing the PHY model
PHY_CTRL_FULLLD   = 0x0100
Total              = 0x6100
```

Fix — remove 100MB from PHY loopback write too:

```
c

mdio_write(PHY_ADDR, PHY_CTRL,
            PHY_CTRL_LOOPBACK | PHY_CTRL_FULLLD); // 0x4100 only
```

Check 2 — Is there a gap between Test 3 and Test 4?

After writing PHY loopback, the PHY simulation model may need time to settle before it starts echoing. The original `delay(10000)` was handling this. Your fix removed that delay:

```
c

// Your current Test 3 fix
mdio_write(PHY_ADDR, PHY_CTRL, PHY_CTRL_LOOPBACK | PHY_CTRL_FULLLD);
while(RD(GRETH_MDIO) & MDIO_BUSY); // waits for MDIO only

// What you might need — wait for PHY to actually enter loopback
while(RD(GRETH_MDIO) & MDIO_BUSY); // wait for MDIO transaction
delay(50000);                       // then give PHY time to switch mode
```

Check 3 — Verify PHY loopback readback value on waveform

In Test 3 you read back PHY CTRL after writing. Check on transcript:

```
PHY CTRL after write = 0x????
```

It should show `0x4100` (loopback + full duplex) or `0x6100` (loopback + 100MB + full duplex). If it shows anything else the write did not stick properly.

Most Probable Fix Sequence

Based on everything — RXD flat, RX_DV flat, original systest passes — here is what to do right now in order:

Change 1 — Remove 100Mbit everywhere:

```
c
```

```
// Test 1
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD);

// Test 3 PHY write
mdio_write(PHY_ADDR, PHY_CTRL, PHY_CTRL_LOOPBACK | PHY_CTRL_FULLD);

// Test 4
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD | CTRL_PROM);

// Test 5
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD | CTRL_PROM);
```

Change 2 — Add settle delay after PHY loopback write in Test 3:

```
c

while(RD(GRETH_MDIO) & MDIO_BUSY);
delay(50000); // give PHY model time to enter loopback mode
```

Change 3 — Verify descriptor order in Test 4:

```
c

txd[0].addr = (unsigned int)txbuf[0];
txd[0].ctrl = DESC_EN | DESC_WRAP | FRAME_SIZE;
rx[0].addr = (unsigned int)rxbuf[0];
rx[0].ctrl = DESC_EN | DESC_WRAP | 128;

WR(GRETH_TXDESC, (unsigned int)txd); // before CTRL
WR(GRETH_RXDESC, (unsigned int)rx); // before CTRL
WR(GRETH_CTRL, CTRL_TXEN | CTRL_RXEN | CTRL_FULLD | CTRL_PROM);
```

After Making Changes — What to Look For

Run simulation again and check waveform in this order:

Step 1: Is TX_EN pulsing now?

NO → descriptor or speed issue still present

YES → go to step 2

Step 2: Is RX_DV pulsing after TX_EN goes low?

NO → PHY loopback still not working, increase delay(50000) more

YES → go to step 3

Step 3: Is rxbuf showing non-zero values?

NO → AHB DMA write path broken

YES → check if values match txbuf → test should PASS

Tell me what TX_EN looks like on the waveform — that one signal will tell us exactly where in the chain the problem is sitting.

TX_EN pulsing

TX_EN pulsing means **GRETH is transmitting correctly** — so the problem is specifically in the **PHY loopback not echoing back**.

The chain breaks exactly here:

- ✓ LEON3 → writes TX descriptor
- ✓ GRETH fetches frame over AHB DMA
- ✓ GRETH drives TXD + TX_EN on MII
- ✗ PHY receives it but does NOT echo back
- ✗ RXD + RX_DV stay flat
- ✗ rxbuf stays all zeros

Check These on Waveform Right Now

Check 1 — Is TXD actually carrying correct data?

Zoom into the TX_EN pulse and look at TXD at the same time:

TX_EN ———┐ ———┐

TXD ———|F|F|F|F|F|F|0|2|... ← should see FF FF FF FF FF FF 02...

↑ destination MAC bytes

If TXD is toggling with correct values — frame is going out fine and PHY is the problem.

Check 2 — Look at MDC and MDIO during Test 3

Zoom into Test 3 on waveform and check:

MDC  ← should be toggling during MDIO write
 MDIO  ← should carry loopback command bits

If MDC is flat during Test 3 — the MDIO write never actually happened and PHY loopback was never enabled.

Check 3 — What does transcript say for Test 3?

Look at your VCS transcript output:

```
--- TEST 3: PHY Loopback Mode ---
PHY CTRL after write = 0x????    ← what value is printed here?
PASS/FAIL: PHY loopback bit confirmed via readback
```

Tell me that value — it tells us exactly what the PHY received.

Most Likely Cause

Since TX_EN pulses but RX_DV is flat, the **PHY loopback mode is not active**. This happens because of one of these:

Reason 1 — 100Mbit bit confusing the PHY model

Your Test 3 writes:

```
c
mdio_write(PHY_ADDR, PHY_CTRL,
            PHY_CTRL_LOOPBACK | PHY_CTRL_100MB | PHY_CTRL_FULLD);
// = 0x4000 | 0x2000 | 0x0100 = 0x6100
```

The GRLIB `eth_phy` simulation model may **ignore loopback when 100MB is set** if it only models 10Mbit. Try:

```
c
// Remove PHY_CTRL_100MB
mdio_write(PHY_ADDR, PHY_CTRL,
            PHY_CTRL_LOOPBACK | PHY_CTRL_FULLD);
// = 0x4000 | 0x0100 = 0x4100
```

Reason 2 — PHY model needs more settle time

After MDIO write, PHY simulation model needs simulation time to switch into loopback mode. Your current fix:

c

```
while(RD(GRETH_MDIO) & MDIO_BUSY); // only waits for MDIO transaction
```

This exits as soon as MDIO bus is free — but the PHY model internally may need more clock cycles to actually activate loopback. Add a delay after:

c

```
while(RD(GRETH_MDIO) & MDIO_BUSY);  
delay(100000); // give PHY model time to activate loopback
```

Reason 3 — PHY model requires a specific loopback sequence

Some GRLIB PHY models require you to **reset the PHY first** before setting loopback:

c

```
// Reset PHY first  
mdio_write(PHY_ADDR, PHY_CTRL, PHY_CTRL_RESET);  
while(RD(GRETH_MDIO) & MDIO_BUSY);  
delay(50000); // wait for reset to complete  
  
// Then set loopback  
mdio_write(PHY_ADDR, PHY_CTRL,  
           PHY_CTRL_LOOPBACK | PHY_CTRL_FULLD);  
while(RD(GRETH_MDIO) & MDIO_BUSY);  
delay(100000); // wait for loopback to activate
```

The Fastest Way to Confirm

Look at your VCS transcript and tell me:

1. What value is printed for PHY CTRL after write in Test 3?

PHY CTRL after write = 0x????

2. Did Test 3 PASS or FAIL?

PASS: PHY loopback bit confirmed via readback
FAIL: PHY loopback bit confirmed via readback

3. On waveform — does MDC toggle during Test 3?

MDC toggling = MDIO write happened
MDC flat = MDIO write never reached PHY

These three answers will tell us **exactly** which reason above is causing the problem and give us the precise fix.

PASS: PHY loopback bit confirmed via readback and MDC toggling = MDIO write happened
PHY CTRL after write = 0x???? I don't remember that

OK so PHY loopback **is confirmed set** and MDIO write **did happen** — so the issue is not the loopback configuration itself.

This means the problem is:

- ✓ PHY loopback bit is SET correctly
- ✓ MDIO write happened
- ✓ TX_EN pulsing — frame going out
- ✗ PHY not echoing back — RXD/RX_DV flat

The Most Likely Reason Now

The PHY model is not getting enough simulation time to activate loopback before the frame arrives.

Think of it like this:

Test 3: writes loopback bit —————
↓ PHY needs time to switch
Test 4: immediately sends frame —————→ PHY ← frame arrives BEFORE loopback is active!
so PHY drops it

The MDIO write completed but the **PHY simulation model internally hasn't finished switching to loopback mode yet** when your frame arrives.

The Fix

Add a larger delay in Test 3 **after** your MDIO busy poll:

```
c

static void test_phy_loopback_mode(void)
{
    unsigned int ctrl;
    printf("\n--- TEST 3: PHY Loopback Mode ---\n");

    mdio_write(PHY_ADDR, PHY_CTRL,
               PHY_CTRL_LOOPBACK | PHY_CTRL_FULLD);

    // Wait for MDIO transaction to finish
    while(RD(GRETH_MDIO) & MDIO_BUSY);

    // Give PHY model time to actually activate loopback internally
    delay(200000);    // increase this if still not working

    ctrl = mdio_read(PHY_ADDR, PHY_CTRL);
    printf("    PHY CTRL after write = 0x%04X\n", ctrl);
    check("PHY loopback bit confirmed via readback",
          ctrl & PHY_CTRL_LOOPBACK);
}
```

If That Still Doesn't Fix It

Check what the **original systest.c does for loopback** — since it passes, it must be handling this settle time somehow:

```
bash

grep -A 20 "loopback\|LOOPBACK\|PHY_CTRL" /path/to/original/systest.c
```

Compare that section with your Test 3 and match exactly what it does — the delay value, the sequence, the register values. That will give you the exact working configuration.

Can you share what the original systest.c does for PHY loopback? That will get this solved immediately.

give me a check list

Here's your complete debug checklist based on everything we've discussed:

GRETH Test Debug Checklist

SETUP CHECKS

- greth_test.c copied/linked as systest.c in correct design folder
- make soft ran successfully and generated .srec file
- .srec file renamed/linked as ram.srec (or reference updated)
- make vcs-launch runs without compile errors
- VCS transcript shows test banner printed (proves LEON3 is running)

TEST 1 CHECKS

- PASS: GRETH reset bit self-clears
 - if FAIL: GRETH_BASE address wrong, check config.vhd
- PASS: MAC MSB write-readback
- PASS: MAC LSB write-readback
 - if FAIL: APB write works but read broken, check bridge read path
- PASS: TX enable bit set
- PASS: RX enable bit set
- PASS: Full-duplex bit set
 - if any FAIL: CTRL register not accessible, check APB bridge
- PASS: 100Mbit bit set
 - if FAIL: remove CTRL_100MB, PHY model is 10Mbit only
 - check: grep "100\|speed" original systest.c

TEST 2 CHECKS

- PHY ID1 printed and not 0xFFFF
- PHY ID2 printed and not 0xFFFF
 - if 0xFFFF: MDIO bus not connected, check MDC/MDIO signals in testbench
- PHY STATUS printed and not 0xDEAD
 - if 0xDEAD: MDIO busy never cleared, mdio_wait() timed out

- PASS: No MDIO link failure
 - if FAIL: speed mismatch between GRETH and PHY
 - fix: remove CTRL_100MB from Test 1 first, then recheck this

TEST 3 CHECKS

- MDC toggling on waveform during Test 3
 - if flat: MDIO write never happened
- PHY CTRL after write printed
 - value should be 0x4100 (loopback+fullduplex, no 100MB)
 - or 0x6100 (loopback+100MB+fullduplex) if 100MB is supported
- PASS: PHY loopback bit confirmed via readback
 - if FAIL: replace delay(10000) with while(RD(GRETH_MDIO) & MDIO_BUSY)
- Settle delay added after MDIO busy poll
 - delay(200000) minimum after while(MDIO_BUSY)
 - if Test 4 RX still fails, increase to delay(500000)
- Check original systest.c loopback sequence and match it
 - grep -A 20 "loopback" /path/to/original/systest.c

TEST 4 CHECKS

- Descriptor setup order is correct:
 1. fill_eth_header(txbuf[0], 0) ← build frame first
 2. txd[0].addr = txbuf[0] ← point descriptor to buffer
 3. txd[0].ctrl = DESC_EN|DESC_WRAP|FRAME_SIZE
 4. rxd[0].addr = rxbuf[0]
 5. rxd[0].ctrl = DESC_EN|DESC_WRAP|128
 6. WR(GRETH_TXDESC, txd) ← set descriptor BEFORE ctrl
 7. WR(GRETH_RXDESC, rxd) ← set descriptor BEFORE ctrl
 8. WR(GRETH_CTRL, ...) ← enable LAST
- CTRL written without CTRL_100MB
 - WR(GRETH_CTRL, CTRL_TXEN|CTRL_RXEN|CTRL_FULLD|CTRL_PROM)
- On waveform – TX_EN pulses
 - if flat: DMA path broken or descriptor not seen by GRETH

- On waveform – TXD carries correct data
 - first 6 bytes should be FF FF FF FF FF FF
 - next 6 bytes should be 02 00 00 00 00 08
 - bytes 12-13 should be 08 00
 - bytes 14+ should count upward from 0x0E
- PASS: Single frame TX completed
 - if FAIL: increase TIMEOUT value in code
- PASS: No TX errors
 - if FAIL: AHB error on DMA fetch, check bridge reverse path
 - look for HRESP=ERROR on waveform
- On waveform – RX_DV pulses after TX_EN goes low
 - if flat: PHY loopback not active, increase settle delay in Test 3
- On waveform – RXD carries same data as TXD
 - if flat/zero: frame not reaching GRETH RX path
 - if garbled: speed mismatch, remove CTRL_100MB
- rxbuf not all zeros
 - if all zeros: RX_DV flat = PHY loopback issue
 - RX_DV high = AHB DMA write path broken
- PASS: Single frame RX completed
 - if FAIL after RX_DV seen: RX descriptor issue
- PASS: No RX errors
 - if FAIL: AHB error on DMA write to rxbuf, check bridge
- PASS: Loopback frame payload matches
 - if FAIL: compare txbuf vs rxbuf byte by byte in transcript
 - mismatch at byte 0-5: destination MAC wrong
 - mismatch at byte 6-11: source MAC wrong
 - mismatch at byte 14+: payload corrupted in transit

TEST 5 CHECKS

- All 4 TX descriptors set up before enabling CTRL
- WRAP bit set only on last descriptor ($i == \text{NUM_FRAMES}-1$)
- All 4 RX descriptors set up with correct buffer addresses
- On waveform – TX_EN pulses 4 times
 - if only 1 pulse: descriptor ring WRAP issue

- PASS: All TX frames sent
 - if FAIL: not all DESC_EN bits cleared by GRETH
- PASS: No TX errors (stress)
 - if FAIL: AHB error during multi-frame DMA
- On waveform – RX_DV pulses 4 times
 - if less than 4: some frames not looping back
- PASS: All RX frames received
 - if FAIL: RX descriptor ring not set up correctly
- No "Frame N still owned by HW" messages
 - if seen: GRETH never processed that descriptor
 - fix: check WRAP bit, check CTRL_100MB removed, add settle delay
- PASS: All frame payloads match (stress)
 - if FAIL: note which frame and which byte failed
 - frame 0 payload starts 0x0E
 - frame 1 payload starts 0x1E
 - frame 2 payload starts 0x2E
 - frame 3 payload starts 0x3E

WAVEFORM SIGNAL CHECKLIST

- PSEL/PENABLE toggling → APB bus active
- PADDR showing correct offsets → 0x00 CTRL, 0x08 MACMSB, 0x10 MDIO
- MDC toggling in Tests 2 & 3 → MDIO bus active
- TX_EN pulsing in Tests 4 & 5 → GRETH transmitting
- TXD shows FF FF FF FF FF FF... → correct frame going out
- RX_DV pulsing after TX_EN → PHY loopback echoing
- RXD matches TXD data → clean loopback
- No X values on any data bus → no uninitialized signals
- HRESP never ERROR → AHB bridge healthy
- DESC_EN cleared after TX/RX → DMA completed successfully

FINAL RESULT CHECKS

- Passed : 22
- Failed : 0
- *** GRETH ETHERNET TEST PASSED ***

- If passed – srec is valid for your external GRETH bridge setup
- If any failures remain – fix top to bottom, Test 1 first always

CURRENT STATUS (your session so far)

- ✓ Test 1 – mostly passing (remove CTRL_100MB)
- ✓ Test 2 – will fix after CTRL_100MB removed
- ✓ Test 3 – fixed with MDIO busy poll (add delay(200000) after)
- ⚠ Test 4 – TX working, RX flat (increase PHY settle delay)
- ⚠ Test 5 – frames owned by HW (will fix after Test 4 RX works)

Fix CTRL_100MB first — it is the single change most likely to unblock Tests 4 and 5 together.

